

CSE 221

$$= 31 \times 7 + 30 \times 4 + 28 = 3965$$

$$(31 \times 7) + (30 \times 4)$$

Handwritten calculations and notes on the right side of the page, including numbers like 302, 500, 550, 410, 87, 31, 7, 217, 30, 4, 120, and some scribbles.

CSE 221

- * 10^8 (or) 10^9 range as loop run করতে - মাত্র - 1 second.
- * 1 GHz মানে কী? # কম্পিউটার-এর ঘড়ি যোগ করতে মাত্র.
- * $O(n^2)$ → Nested loop time complexity.

Big O

→ Order of

→ time complexity

* অনেকগুলো থাকলে যেটা বড় বুঝানো যেটা Big O notation এ

$$O(\max(N, M))$$

for i in range(N):

for i in range(M):

* constant কিছু range এ থাকলে যেটা - $O(1)$ →

* constant এর টার থাকা/না থাকা matter করে না.

for i in range $(1, N)$:
 for j in range (i, N) :

Time complexity $\rightarrow O(n^2)$

i	j
1	N
2	$N-1$
3	$N-2$
4	$N-3$
$\vdots$	

Account constant $2(N-1)$
 $0 - 2(N-1) - \dots - 1$
 Get total $\rightarrow$
 $O(n^2) - 2(N-1) - \dots - 1$

Square root of N

* Order of square root of $N \rightarrow$ Time complexity of checking prime. $\frac{N}{2}$ is a power that is why it can not be written as $O(N)$

* $\log(n)$

1. for i in range $(1, n, n/2)$: $\rightarrow$ Get n by $n/2$

2. for j in range $(1, n, \frac{n}{2})$: $\rightarrow$ Get n first last $n/2$ $\rightarrow$ Get $n/2$ $\rightarrow$ Get $n/2$

* $1 + 2 + 3 + \dots + N \rightarrow$ Sum $O(N)$
 $\rightarrow$ Get n first $\rightarrow$ Get n $O(1)$

